Limits & Limits with Infinity
(Sections 2.2, 2.6)

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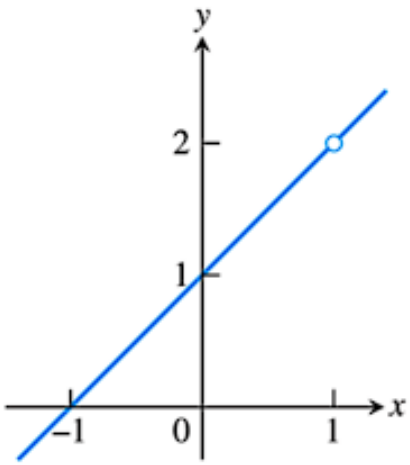
MATH 2413

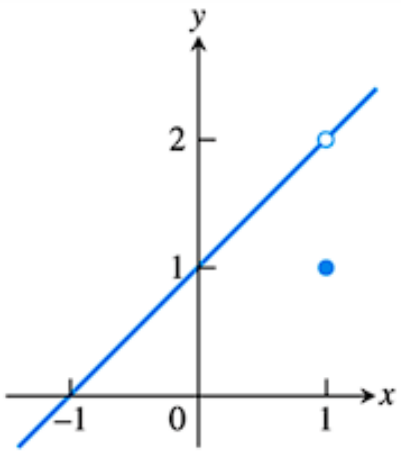
Basics of Limits

Definition

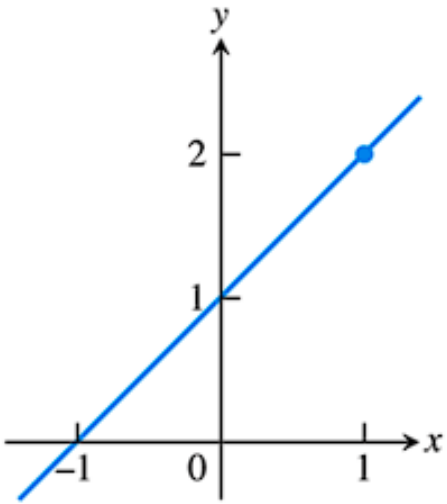
As opposed to a function evaluation, a **limit** explores how a function behaves as x (or any other variable) *approaches* a particular value.



(a) $f(x) = \dfrac{x^2 - 1}{x - 1}$



(b) $g(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \neq 1 \\ 1, & x = 1 \end{cases}$

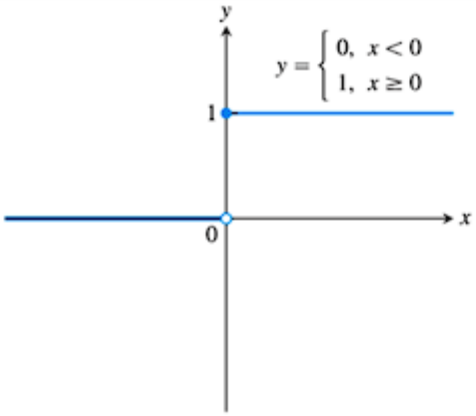


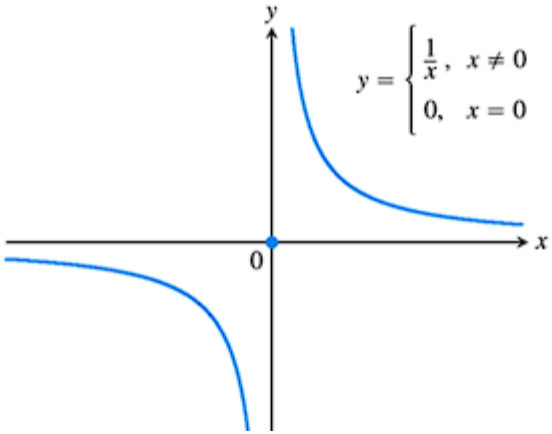
(c) $h(x) = x + 1$

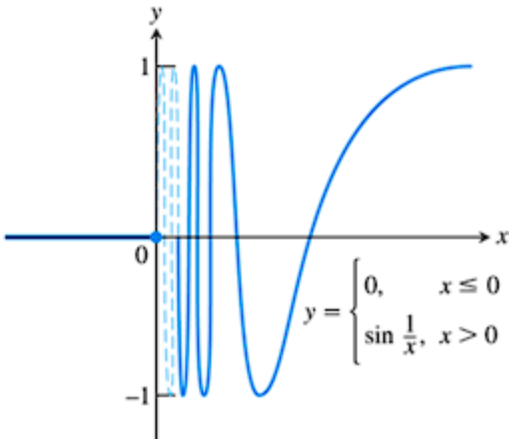
Notation / When Limits Do Not Exist

We denote a limit as $\lim_{x \to c} f(x)$.

If approaching from the left and right sides do not each yield the same value, we say *the limit does not exist*.



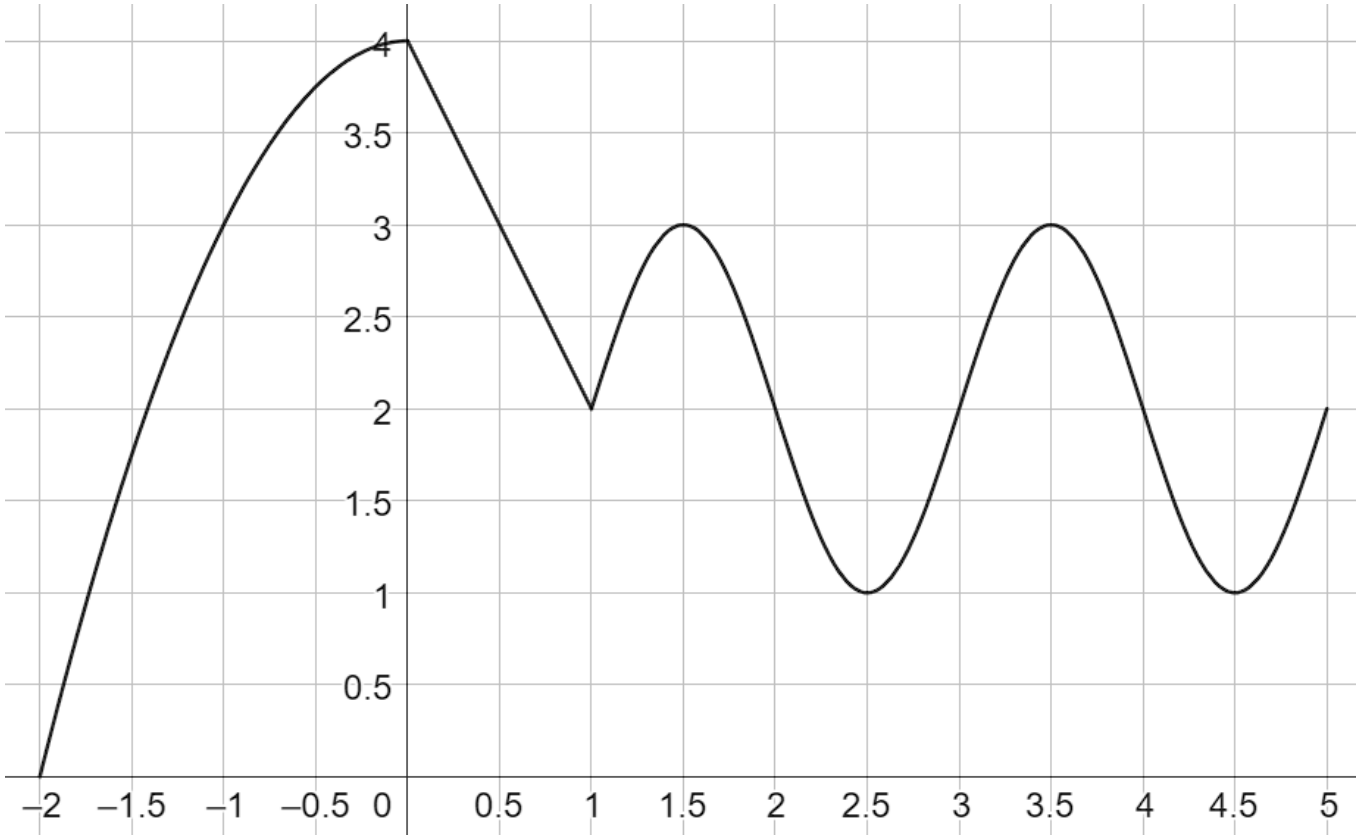




Examples

Example

Compute $\lim_{x\to -2} f(x)$, $\lim_{x\to 0} f(x)$, $\lim_{x\to 1} f(x)$, and $\lim_{x\to 4} f(x)$.



Limit Rules

Theorem

Suppose $\lim_{x \to c} f(x) = L$ *and* $\lim_{x \to c} g(x) = M$*. Then,*

- $\lim_{x \to c} f(x) \pm g(x) = L \pm M$ *(Sum/Difference Rule)*
- $\lim_{x \to c} k \cdot f(x) = k \cdot L$ *(Constant Multiple Rule)*
- $\lim_{x \to c} f(x) \cdot g(x) = L \cdot M$ *(Product Rule)*
- $\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{L}{M}$ *(Quotient Rule)*
- $\lim_{x \to c} f(x)^n = L^n$ *(Power Rule)*
- $\lim_{x \to c} \sqrt[n]{f(x)} = \sqrt[n]{L}$ *(Root Rule)*

Algebraic Examples

Example

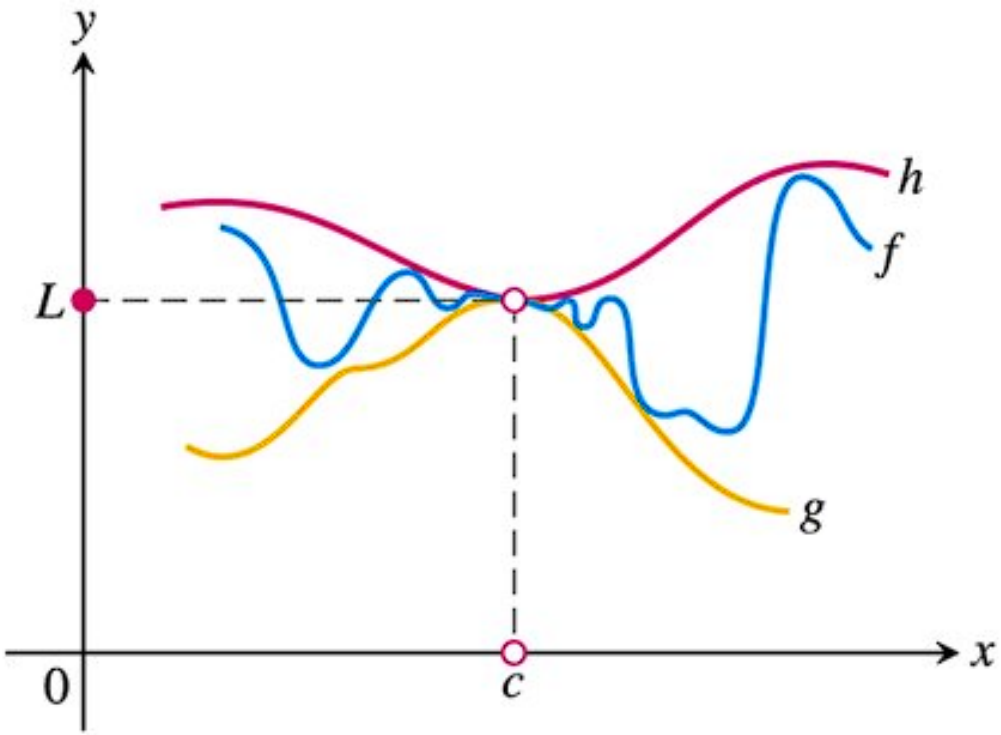
Compute the following limits:

- $\lim_{x \to -1} \frac{x^3 - 2x + 1}{3x + 5}$
- $\lim_{x \to 2} \frac{x^3 - 8}{7x - 14}$
- $\lim_{x \to 9} \frac{\sqrt{x} - 3}{x^2 - 81}$
- $\lim_{x \to -1} \frac{\frac{1}{x} + \frac{1}{x^2}}{x + 1}$

Sandwich Theorem

Theorem

Suppose that $g(x) \leq f(x) \leq h(x)$ *for all* x *near* c *except possibly at* $x = c$ *itself. If* $\lim_{x \to c} g(x) = \lim_{x \to c} h(x) = L$*, then* $\lim_{x \to c} f(x) = L$*.*



Example

Example

Suppose $1 - x \leq f(x) \leq 1 - x^2$ for all $x \in (-1, 1)$. Find $\lim_{x \to 0} f(x)$.

A Famous Example

Example

Compute $\lim_{\theta \to 0} \frac{\sin(\theta)}{\theta}$.

Basic Infinite Limit Examples

Example

Compute the following.

- $\lim_{x \to 0} \frac{1}{x^2}$
- $\lim_{x \to 0} \frac{-1}{x^2}$
- $\lim_{x \to \infty} \frac{1}{x^2}$
- $\lim_{x \to -\infty} \frac{1}{x^2}$

One Sided Limit Notation

Definition

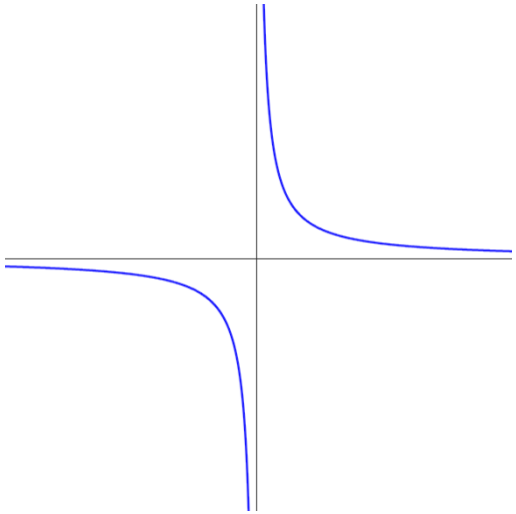
A **one-sided limit** is much like a standard (two-sided) limit, only taken from one direction.

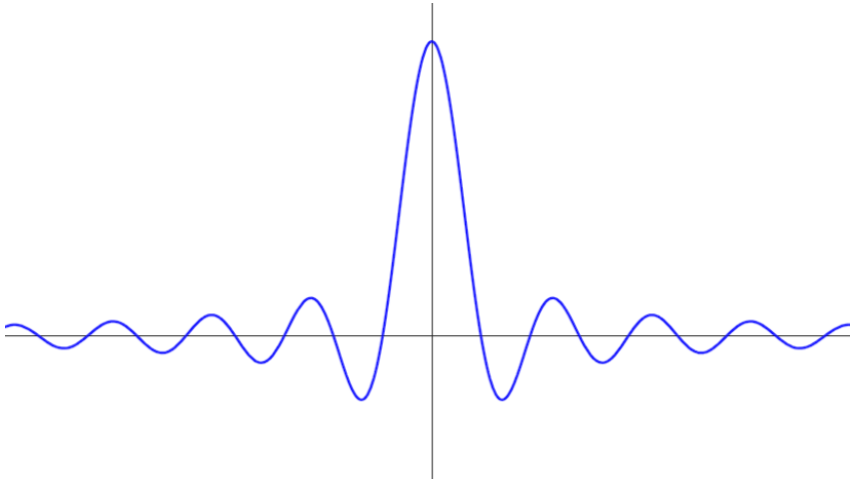
- $\lim_{x \to c^-} f(x)$ denotes a limit coming from the left side (i.e. numbers smaller than x).
- $\lim_{x \to c^+} f(x)$ denotes a limit coming from the right side (i.e. numbers bigger than x).

Asymptotes

Definition

- A function has a **horizontal asymptote** at $y = L$ if and only if $\lim_{x \to \pm\infty} f(x) = L$.
- A function has a **vertical asymptote** at $x = c$ if and only if $\lim_{x \to c^{\pm}} f(x) = \pm\infty$.





Examples

Example

Find all asymptotes of the following functions.

- $f(x) = \frac{x}{x+1}$
- $g(x) = \frac{x^2+2x-3}{x^2-9}$